

3. Requirements Gathering

3.1 Stakeholder analysis

Stakeholder	Needs
Building Admin	manage apartments, residents, approvals, meters, billing, reports
Residents	view costs, receive alerts, access reports, transparency
Maintenance Staff	receive emergency/issue notifications
Building Owner / Committee	oversight, summaries, accountability
System Developer/Admin	maintain database, security, backups

3.2 User stories

- As a building admin, I want to create and manage buildings and apartments so that resident records are organized.
- As a resident, I want to join a building using an access code so that I can access my apartment information.
- As an admin, I want to approve or reject join requests so that unauthorized users cannot access building data.
- As an admin, I want to enter monthly global electricity and water meter readings so that utility costs can be calculated.
- As an admin, I want the system to distribute utility costs across apartments so that bills are fair and transparent.
- As a resident, I want to receive alerts for emergencies or high consumption so that I can respond quickly.
- As a resident, I want to download or view monthly PDF reports so that I can review charges and consumption.
- As a user, I want the app to work with weak connectivity so that I can still access important information.

3.3 Use cases

Main use cases:

1. Register/login
2. Create building
3. Add apartment
4. Generate access code
5. Join building via access code
6. Approve resident request
7. Record global meter reading
8. Configure cost distribution
9. Generate alerts
10. Produce monthly PDF report
11. Sync offline data

3.4 Functional requirements

- The system shall allow admins to create and manage building profiles.
- The system shall allow admins to register apartments under a building.
- The system shall allow residents to request joining a building using an access code.
- The system shall require admin approval before granting resident access.
- The system shall store global electricity and water meter readings.
- The system shall calculate and distribute costs to apartments according to selected rules.
- The system shall send or display alerts categorized as normal, emergency, or consumption alerts.
- The system shall generate monthly PDF reports for each apartment/building.
- The system shall cache selected data locally using SQLite.
- The system shall synchronize local and remote data when internet access becomes available.

3.5 Non-functional requirements

- Performance: common screens should load quickly.
- Reliability: sync conflicts should be minimized and recoverable.
- Security: authenticated access, role-based authorization, secure storage.
- Usability: simple Arabic-friendly mobile UI.
- Scalability: support multiple buildings and growing resident count.
- Maintainability: modular code and documented architecture.
- Availability: core cached data accessible offline.
- Portability: mobile app runs on Android initially, optionally iOS later.